Ranadeep Daw

Curriculum Vitae

Division of Translational Toxicology |
Biostatistics & Computational Biology Branch
National Institute of Environmental Health Sciences
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Current Position

2023– Postdoctoral fellow, Division of Translational Toxicology, Biostatistics & Computational Biology Branch, National Institute of Environmental Health Sciences, Durham, NC 27709.

Research Interests

Spatiotemporal modeling, Exposure models, Environmental risk mapping, Low-rank methodologies, Data reduction, Random neural networks, Computational statistics, Bayesian analysis

Education

2018–2023 **Ph.D. in Statistics**, *University of Missouri - Columbia*

GPA: 3.878

Advisor: Dr. Christopher K. Wikle

Thesis topic: Methodologies for low-rank analysis and regionalization for multi-scale spatial datasets

2014–2016 **Master of Statistics (M. Stat.)**, *Indian Statistical Institute*, Kolkata

2011–2014 **Bachelor of Statistics (B. Stat.)**, *Indian Statistical Institute*, Kolkata

Work Experience

Jul 2023– **Postdoctoral fellow**, *National Institute of Environmental Health Sciences*, Durham, NC
Working on spatiotemporal exposure modeling and supervised environmental risk modeling.

Aug 2018– Jul 2019 **Intern**, *Missouri Department of Conservation*, Columbia, MO
Served as part of my research assistantship. Worked on species distribution models, small area estimation, and bootstrapping.

Jun 2016– Dec 2017 **Data Scientist**, *Deloitte US India*, Hyderabad, India
Addressed statistical queries in various business-related sectors and implement them in user-friendly tools for third-party use. Worked on real-world data from the insurance, banking, medical, and retail sectors.

May 2015– Jul 2015 **Intern**, *General Electric*, Bangalore, India
Implemented fast heuristic time series algorithms.

May 2014– Jul 2014 **Intern**, *FinIQ*, Pune, India
Visual Basic-based automation of financial algorithms in Excel.

Research Projects

Accepted Papers

1 Schliep, E. M., Wikle, C. K., **Daw, R.** (2023). Correcting for informative sampling in spatial covariance estimation and kriging predictions. *Journal of Geographical Systems*, 1–27.

2 **Daw, R.**, Wikle, C. K. (2022). Supervised spatial regionalization using the Karhunen-Loève Expansion and minimum spanning tree. *Journal of Data Science* 20 *no.* 4. 566–584, *DOI 10.6339/22-JDS1077*.

3 **Daw, R.**, Wikle, C. K (2022). REDS: Random Ensemble Deep Spatial prediction. *Environmetrics, e2780,* 1180–4009, *DOI 10.1002/env.2780*.

4 **Daw, R.**, Simpson, M., Wikle, C. K., Holan, S. H., Bradley, J. R. (2022). An overview of univariate and multivariate Karhunen Loève expansions in Statistics. *Journal of the Indian Society for Probability and Statistics*, 1–42.

5 Chakraborty, S., Menifield, C. E., **Daw, R.** (2022). Impact of Stand Your Ground, Background Checks, and Conceal and Carry Laws on homicide rates in the US. *Journal of Public Management and Social Policy*.

6 Yeasmin, F., **Daw, R.**, Chakraborty, B., Gupta, A., Bhattacharya, S., Chakraborty, B. (2021). A new growth rate measure in identifying extended Gompertz growth curve and development of goodness-of-fit test. *Calcutta Statistical Association Bulletin*, Volume 73, 127–145.

Submitted Papers

7 **Daw, R.**, Wikle, C.K., Bradley, J. R., Holan S. H. (2023). MVCAGE: A criterion for multivariate regionalization of spatial data.

Ongoing Papers

8 **Daw, R.**, Messier P. K. (2024+). SBoost: A spatially-aware boosting model for global high-resolutional imputation of NO_2.

9 **Daw, R.**, Messier P. K. (2024+). A supervised dimension reduction methodology for geospatial conditional risk modeling.

10 **Daw, R.**, Wikle, C. K. (2024+). Application of coresets in optimal spatial data reduction.

11 Yoo M., Grieshop N., **Daw, R.**, Wikle, C. K. (2024+). A comprehensive review of uncertainty quantification methodologies for reservoir computing models.

Conferences

Invited Talks

2021 **JSM**, *(Joint Statistical Meetings)*, Application of coresets in Spatial Modeling.

Contributed Talks

2021 **IISA Conference**, *(International Indian Statistical Association)*, Application of coresets in Spatial Modeling.

2020 **JSM**, *(Joint Statistical Meetings)*, Uncertainty Quantification and Inference for Spatiotemporal Forecasting via Echo State Mixture Density Networks with Relevance Propagation.

Poster Presentations

2024 **Theory and Foundations**, *Florida State University*, SBoost: A spatially-aware Boosting Model for Large Spatial Prediction.

2024 **Science Day**, *NIEHS (National Institute of Environmental Health Sciences)*, SBoost: A spatially-aware Boosting Model for Large Spatial Prediction.

2022 **Applied Statistics Symposium**, *ICSA (International Chinese Statistical Association)*, REDS: Random Ensemble Deep Spatial prediction.

2019 **Deep Learning Program Opening Workshop**, *SAMSI (Statistical and Applied Mathematical Sciences Institute)*, Deep Neural Network in Cusp Catastrophe Model

2018 **Innovations in Design, Analysis, and Dissemination**, *Frontiers in Biostatistical Methods*, Does Changing Handgun Laws Affect Crime Rates?

Teaching Experience

Spring 2020 **Graduate Instructor**, *Stat 3500*, Probability and Statistics - II
- *Topics*: ANOVA, Linear and Logistic Regression, Nonparametric Statistics, Introduction to R.
- *Responsibilities*: Instruction and evaluation.
- *Mode*: Both online and classroom.
- *Special Mention*: **Data Analysis project** as Final examination.

Fall 2019 **Graduate Instructor**, *Stat 3500*, Probability and Statistics - II
- *Topics*: ANOVA, Linear and Logistic Regression, Nonparametric Statistics, Introduction to R.
- *Responsibilities*: Instruction and evaluation.
- *Mode*: Classroom.

Spring 2018 **Teaching Assistant**, Evaluation of examination and homework of various graduate and undergraduate Statistics courses.

Service & Membership

Peer Review	Reviewer for *Spatial Statistics*, *Signal Processing*.
Professional Membership	Member of *American Statistical Association*.

Technical Expertise

Computing	MATLAB, `Python`, `SAS`, `R`.
Programming	`C`, `C++`, LaTeX.
Visualization	`Tableau`, `HTML`, `JavaScript`, `JavaScript-D3`.
Cloud	`Lewis` cluster, `Nautilus`, `Google colab`.

Academic Awards

2011–2016	Received the **INSPIRE** scholarship administered by the Department of Science and Technology of the Government of India.
2010	Cleared the **Regional Mathematical Olympiad** and qualified for the national level examination.

Miscellaneous

Leadership Experience

2020	Organized spatiotemporal research meetings at University of Missouri – Columbia.
2019–20	Committee member of the Statistics graduate students association at University of Missouri – Columbia.
2015	General Secretary of the annual techno-cultural and sports festival at the Indian Statistical Institute, Kolkata.
2014–15	Student Representative of the boys' hostel of Indian Statistical Institute, Kolkata.
2014	Treasurer of the annual college techno-cultural and sports festival at the Indian Statistical Institute, Kolkata.
2012–14	Member of food, sports, and cultural committees at the boys' hostel of Indian Statistical Institute, Kolkata.

Mentoring Experience

1. Worked as consultant in the Data Fest, a data analysis workshop organized by the Department of Statistics, University of Missouri.
2. Responsible for mentoring two newly-hired employees at Deloitte. Worked on a coding project with one mentee.

Extracurricular Activities

Play contract bridge, soccer.

Languages Known

English, Bengali (mother tongue), Hindi.

I hereby do certify that all the above information is true to the best of my knowledge.
Ranadeep Daw
June 15, 2024